Multi-Site Brain MRI Landmark Detection

A Six-Approach Pipeline Evaluation

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Abstract

We develop and evaluate a multi-site brain MRI landmark detection pipeline that predicts 51 anatomical landmarks for subjects across six simulated acquisition sites. Starting from a simple per-site mean baseline (4.59 mm), we implement five progressively sophisticated approaches including rigid registration, posterior-guided heuristic search, 3D convolutional neural network (CNN) regression, multi-atlas affine registration, and a BCD-inspired linear least-squares (LLS) model from tissue posteriors. The CNN achieves the best internal accuracy at **4.54 mm** mean Euclidean error on a 20% per-site held-out evaluation set, and is selected for final submission of predictions for all 162 unlabeled subjects. Key findings: (i) rigid ACPC-space registration is a null result relative to the mean baseline; (ii) posterior-weighted centroids are unreliable for sub-structure boundaries; (iii) coordinate normalisation is critical for regression heads predicting targets far from the origin; (iv) orbital landmarks (LE, RE) remain the dominant error source at 11–13 mm across all approaches.

1 Site Characteristics

Data comprise 642 labeled subjects (approximately 107–108 per site) and 162 unlabeled subjects (27 per site) across six sites. Each subject provides a T1 NIfTI volume, an optional T2, six tissue posterior maps (WM, GM, CSF, Ventricle Bodies, Globus Pallidus, Background), and — for labeled subjects — 51 ground-truth landmarks in RAS physical space (mm).

Orientation. NIfTI direction cosines are the identity matrix for every subject in every site, confirmed via `SimpleITK.GetDirection()` on all 642 T1 volumes. All images are in RAS orientation; there are no rotational misalignments between or within sites. This simplifies the physical-to-voxel mapping to a translate-then-scale operation (no direction matrix inverse required).

Voxel spacing. Spacing is constant within each site (intra-site standard deviation ≈ 0) but varies between sites from roughly 0.92 mm to 1.10 mm isotropic. See Table 2 in Appendix Appendix A for per-site values.

Physical origin. The physical origin varies within each site ($\sigma \approx 3$ mm per axis), reflecting different scanner field-of-view placements. Critically, the AC landmark position in scanner space varies by $\sigma \approx 17$ mm per axis for sites B–F, but is always $(0, 0, 0)$ mm for siteA. This reveals that siteA data were pre-aligned to ACPC space before being placed in the enclave, while sites B–F represent native scanner coordinates. The translational scatter is *not* accompanied by rotational scatter — brains are already in RAS orientation in all sites.

Landmark consistency. All 51 landmark labels are identical across all six sites. The AC–PC distance is 27 mm ($\sigma \approx 1$ mm) across all sites, confirming anatomical consistency of the reference landmarks and their suitability as an ACPC-alignment anchor.

Design implication. The principal challenge of cross-site generalization is handling translational offsets and mild spacing variation, *not* arbitrary orientations. A detector operating in ACPC space (where AC is forced to the origin) is therefore immediately site-agnostic, provided the ACPC

transform is estimable.

2 Design Decisions

Six approaches were implemented and evaluated using a deterministic 80/20 labeled split (first 20% of subjects per site as holdout), yielding approximately 21 eval subjects per site. Table 1 summarizes all results.

Table 1: Mean Euclidean landmark error (mm) for all six approaches on the 20% per-site held-out evaluation set. CNN is selected for final submission.

Approach	siteA	siteB	siteC	siteD	siteE	siteF	Overall	Selected
1. Per-site mean (ACPC)	4.86	4.34	4.61	5.03	4.13	4.55	4.59	
2. Rigid registration	4.86	4.34	4.61	5.03	4.13	4.55	4.59	
3. Posterior heuristic*	5.13	—	—	—	—	—	5.13*	
4. 3D CNN regression	4.54	4.38	4.64	5.04	4.13	4.53	**4.54**	✓
5. Multi-atlas affine	5.91	5.47	5.56	5.77	5.09	5.27	5.51	
6. BCD-inspired LLS	4.38	4.96	5.08	5.68	4.93	4.82	4.97	

*Applied to siteA only; siteA only score shown.

2.1 Approach 1 — Per-site Mean in ACPC Space (Baseline)

Each labeled subject's landmarks are transformed to ACPC space via a rigid transform computed from that subject's AC, PC, LE, and RE positions. Landmarks are averaged in ACPC space and then mapped back to scanner space using the target subject's predicted ACPC transform. This formulation is correct: averaging in scanner space would be meaningless for sites B–F because the physical origin differs between subjects. Result: **4.59 mm** overall, range 4.13–5.03 mm. Error is dominated by RE (12.4 mm) and LE (11.4 mm); ACPC-defining landmarks are near-perfect by construction (AC 0.00 mm, PC 1.03 mm).

2.2 Approach 2 — Rigid Registration to Per-site ACPC Template (Null Result)

A mean ACPC T1 intensity template was built per site by resampling all labeled subjects onto a fixed $101 \times 101 \times 96$ voxel grid at 2 mm isotropic, then averaging. A six-degree-of-freedom rigid registration (Euler3D, Mattes Mutual Information, multi-resolution $4\times/2\times/1\times$) mapped each unlabeled subject's T1 to this template. The registration transform was then used to propagate the per-site mean ACPC landmark set into the subject's scanner space.

Null result. Approach 2 matches Approach 1 exactly at **4.59 mm**. The explanation is algebraic: the registration refines the subject's ACPC transform estimate, but the propagated landmarks are still the population mean. Since Approach 1 uses the same mean landmarks in ACPC space, the error is identical — rigid registration of a mean cannot reduce anatomical variation around that mean. Improvement would require either subject-specific landmark adaptation (learning from image content) or denser image-to-image correspondences.

The ACPC template and per-subject approximate ACPC transforms (T_{approx}) are re-used as building blocks for Approaches 3, 4, and 6.

2.3 Approach 3 — Posterior-Guided Local Refinement (Negative Result)

For eight corpus callosum sub-structure landmarks (genu, rostrum, inner corpus bilateral, callosum left/right, m_ax_sup/inf), the mean prediction was replaced with a WM-posterior-weighted centroid

within an 8 mm-radius sphere centred on the mean position. Applied to siteA only.

Negative result. Score regressed to 5.13 mm (+0.54 mm), with genu +4.42 mm and rostrum +4.62 mm. Root cause: the WM posterior (P_{WM}) is high throughout all white matter, not specifically at corpus callosum sub-structures. A centroid near genu captures internal capsule, corona radiata, and neighbouring CC sub-regions equally, pushing predictions toward the bulk WM centre of mass rather than toward the anatomically specific boundary.

This failure motivates a supervised approach (Approach 4) and clarifies the conditions under which tissue posteriors are useful: only when the target landmark *coincides* with the tissue's centre of mass (e.g., ventricular body landmarks).

2.4 Approach 4 — 3D CNN Direct Coordinate Regression (Best Result, Submitted)

A 3D CNN was trained to predict all 51 landmark positions from the ACPC-resampled T1 volume. Architecture: five stride-2 double-convolution encoder blocks ($1 \to 24 \to 48 \to 96 \to 192 \to 384$ channels), adaptive average pooling, two fully connected layers ($384 \to 256 \to 153$), Tanh final activation. Input: $1 \times 96 \times 101 \times 101$ normalised ACPC T1. Output: 51×3 coordinates divided by a scale factor of 120 mm, mapping to $[-1, +1]$ consistent with Tanh. Training: all six sites combined (≈516 subjects), 100 epochs, AdamW, cosine learning-rate schedule, smooth-L1 loss, mild intensity jitter and left-right flip augmentation.

Critical fix — coordinate normalisation. The first training run without normalisation yielded 14.10 mm overall error. A regression head initialised near zero predicts all landmarks near the origin; ACPC-defining landmarks (AC at $(0, 0, 0)$) appear correct while peripheral landmarks at ±60–100 mm are catastrophically wrong. Normalising targets to $[-1, +1]$ with a Tanh-bounded output eliminated this failure mode, reducing overall error to **4.54 mm**.

Cross-site pooling is feasible because ACPC space places all subjects in the same coordinate frame regardless of site.

2.5 Approach 5 — Multi-Atlas Affine Registration

Five siteA subjects (atlases) were randomly selected from training data. For each eval/unlabeled subject, the site-specific T1 was affinely registered to each atlas (twelve DOF, Mattes MI), and the atlases' labeled landmarks were transferred to the subject's space via each registration transform. Final predictions were formed by robust voting (median per axis).

Result: **5.51 mm** overall, worse than the mean baseline. Affine registration in scanner space is more difficult than in ACPC space due to the translational scatter in sites B–F, and the multi-atlas voting step adds noise from imperfect registrations. Pre-initialising with the site mean ACPC transform partially mitigated this, but the method could not surpass Approach 1.

2.6 Approach 6 — BCD-Inspired LLS from Tissue Posteriors

This approach re-implements the core step of BRAINSConstellationDetector's [1] linear landmark localisation model using the project's labeled data. Six tissue posterior volumes (WM, GM, CSF, VB, Globus, Background) are sampled via trilinear interpolation at the 51 mean ACPC landmark positions projected back into each subject's scanner space, yielding a 306-dimensional feature vector per subject. Ridge regression maps these features to the 51 ACPC-space landmark coordinates ($\alpha = 10$, solved as $(X^\top X + \alpha I)^{-1} X^\top Y$). Training uses ground-truth ACPC transforms; inference uses $\mathrm{T}_{\mathrm{approx}}$ from the registration template.

Result: **4.97 mm** overall, with siteA achieving 4.38 mm (better than the CNN). Error is highest on siteD (5.68 mm) where feature-sampling uncertainty from T_{approx} degrades the input. The model underperforms the CNN overall because approximate ACPC transforms introduce systematic errors in the feature-sampling stage, particularly for sites B–F.

3 Verification Strategy

Internal evaluation. All approaches are evaluated on a deterministic 20% holdout set (first 20% of subjects per site, sorted by directory name). This produces approximately 21 eval subjects per site (126 total) and is identical across all six approaches, making results directly comparable. The evaluation metric is mean Euclidean error (mm) over all 51 landmarks and all eval subjects.

ACPC geometric verification. Predicted ACPC alignment is verified geometrically on a representative subset:

- $\|\text{AC}\|_2 < 0.1\,\text{mm}$ (AC at origin)
- $|\text{AC}_{SI} - \text{PC}_{SI}| < 0.1\,\text{mm}$ and $|\text{AC}_{LR} - \text{PC}_{LR}| < 0.1\,\text{mm}$
- $|\text{LE}_{SI} - \text{RE}_{SI}| < 0.1\,\text{mm}$

An HTML verification report (`eda_report_acpc_verification.html`) was generated confirming these constraints hold for all labeled training subjects.

Coordinate space sanity check. The instructor hint confirmed that external evaluation expects scanner-space (not ACPC-space) coordinates. After a RAS/LPS coordinate conversion bug was identified and fixed in the SimpleITK `TransformPoint` wrapper (SimpleITK's physical space is LPS; FCSV files use RAS, requiring sign-flips on the x and y axes), all 162 predictions were regenerated. siteB unlabeled AC positions now read $(-33.4, -6.3, -5.3)$ mm rather than $(0, 0, 0)$ mm.

Test suite. 40 unit tests cover core numerical routines (`acpc.py`, `landmarks.py`, `io.py`, round-trip FCSV I/O, error metric symmetry). All tests pass under `pixi run test`.

4 Reproducibility Recipe

A new user on the MIATT Linux workstation in July 2026 can reproduce all results as follows: clone the repository, run `pixi install` to install the fully-pinned conda and PyPI environment from `pixi.lock` (approximately 5 minutes on the first run), verify with `pixi run test` (all 40 tests pass), and then run `pixi run pipeline --approach cnn --output predictions` to regenerate all 162 unlabeled predictions (requires the NVIDIA RTX 4000 Ada GPU and the data enclave mounted at `/nfs/s-l028/scratch/opt/ece5490/MIATTFINALEXAMDATA/`). The CNN model checkpoint, ACPC template images, and per-subject ACPC volume cache are written to `cache/` automatically on the first run; re-runs use the cache and complete in under five minutes. No site-specific configuration is needed — the pipeline discovers sites and subjects dynamically from the data directory structure.

5 Landmark Selection for Non-linear Registration

In a downstream non-linear atlas registration workflow, predicted landmarks serve as correspondence constraints. Selecting landmarks with high predicted accuracy is essential: a 10 mm error on a correspondence point can locally invert the registration [6]. We recommend the following selection criterion based on CNN error statistics (Table 10 in Appendix Appendix C):

Criterion 1 — accuracy threshold. Exclude landmarks with mean error > 5 mm across

sites. This removes RE (12.6 mm), LE (11.2 mm), cortical poles (r_sup_ext 8.2 mm; top_left/right 8.1 mm; l/r_front_pole 6.5–7.6 mm; l/r_occ_pole 6.2–7.0 mm), lateral inner ear (6.6–7.7 mm), and the dens_axis (5.7 mm). The primary driver of high error is proximity to cortical surface boundaries and orbital regions with low T1 contrast.

Criterion 2 — spatial distribution. To constrain a 3D non-linear deformation field, landmarks should span the brain volume. After applying the accuracy threshold, the remaining 36 landmarks cover mid-line (AC, PC, corpus callosum family: rostrum, genu, mid_lat, mid_basel, RP, RP_front), subcortical structures (caudate heads, lateral ventricles, SMV, VN4, optic chiasm, BPons, CM), and a sparser set of lateral/temporal positions (lat_right, lat_left, l/r_temp_pole, l/r_corp, callosum family). This distribution provides adequate anterior-posterior and superior-inferior coverage.

Criterion 3 — anatomical distinctiveness. Landmarks that coincide with tissue boundaries (e.g., caudate head, optic chiasm) or ventricular structures offer high local contrast for registration validation. Landmarks on continuous smooth surfaces (cortical poles) should be de-weighted even when their predicted position is accepted, as registration optimisers may slide along such surfaces.

Recommendation. Use the 26 landmarks with CNN mean error ≤ 4 mm as hard constraints and the remaining 10 (error 4–5 mm) as soft constraints with reduced weight. Exclude all 15 landmarks with error > 5 mm entirely.

6 Conclusion

The 3D CNN (Approach 4) achieves the best landmark accuracy at **4.54 mm** on the internal 20% holdout and is submitted as the final prediction. The primary bottleneck across all approaches is the orbital landmarks LE and RE (11–13 mm); incorporating T2 imaging or a dedicated eye-detection step would be the highest-leverage future improvement. The pipeline requires zero code changes for a new site, satisfying the extensibility requirement by design.

References

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Appendix A EDA: Site Characteristics Tables

Table 2: Per-site dataset characteristics confirmed by EDA across all 642 labeled subjects.

Characteristic	**siteA**	**siteB**	**siteC**	**siteD**	**siteE**	**siteF**
Labeled subjects	108	106	107	107	107	107
Unlabeled subjects	27	27	27	27	27	27
Voxel spacing (mm, isotropic approx.)	1.00	1.02	1.01	1.03	0.98	1.03
Intra-site spacing std	0.000	0.000	0.000	0.000	0.000	0.000
Direction cosines	Identity (RAS) — all sites, all subjects					
AC scatter σ (mm/axis)	$\approx$0	$\approx$17	$\approx$17	$\approx$18	$\approx$17	$\approx$17
Pre-ACPC-aligned?	Yes	No	No	No	No	No
AC–PC distance (mean $\pm$ std, mm)	27.0 ± 1.0					
Notable artifacts	—	—	`.DS_Store`	—	—	—

Table 3: Image size range and physical FOV per site (approximate).

Site	**Size range (voxels, LR$\times$AP$\times$SI)**	**Spacing (mm)**	**Approx. FOV (mm)**
siteA	176–200 $\times$ 204–236 $\times$ 168–198	1.00	176–200
siteB	248–271 $\times$ 295–322 $\times$ 274–299	1.02	253–276
siteC	273–298 $\times$ 270–297 $\times$ 248–275	1.01	276–301
siteD	230–255 $\times$ 276–305 $\times$ 266–295	1.03	237–263
siteE	281–305 $\times$ 299–331 $\times$ 236–260	0.98	275–299
siteF	249–270 $\times$ 257–289 $\times$ 269–292	1.03	256–278

Appendix B Full Per-Approach Results

Table 4: Approach 1 — Per-site Mean in ACPC Space.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.86	1.67	10.79
siteB	85	21	4.34	1.18	7.22
siteC	86	21	4.61	1.77	9.48
siteD	86	21	5.03	2.31	11.17
siteE	86	21	4.13	1.40	7.50
siteF	86	21	4.55	1.58	8.95
Overall			**4.59**	1.71	11.17

Table 5: Approach 2 — Rigid Registration to ACPC Template.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.86	1.67	10.79
siteB	85	21	4.34	1.18	7.22
siteC	86	21	4.61	1.77	9.48
siteD	86	21	5.03	2.31	11.17
siteE	86	21	4.13	1.40	7.50
siteF	86	21	4.55	1.58	8.95
Overall			**4.59**	1.71	11.17

Table 6: Approach 3 — Posterior-Guided Local Refinement (siteA only).

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	5.13	1.66	11.07
Overall			**5.13**	1.66	11.07

Table 7: Approach 4 — 3D CNN Direct Coordinate Regression (submitted).

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.54	1.79	10.92
siteB	85	21	4.38	1.14	6.57
siteC	86	21	4.64	1.83	9.56
siteD	86	21	5.04	2.61	11.79
siteE	86	21	4.13	1.17	7.36
siteF	86	21	4.53	1.71	9.88
Overall			**4.54**	1.80	11.79

Table 8: Approach 5 — Multi-Atlas Affine Registration.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	5.91	2.34	13.19
siteB	85	21	5.47	1.27	8.36
siteC	86	21	5.56	2.19	11.02
siteD	86	21	5.77	2.44	12.23
siteE	86	21	5.09	1.73	8.83
siteF	86	21	5.27	1.81	10.77
Overall			**5.51**	2.02	13.19

Table 9: Approach 6 — BCD-Inspired LLS from Tissue Posteriors.

Site	**Train**	**Eval**	**Mean (mm)**	**Std**	**Max**
siteA	87	21	4.38	1.27	7.30
siteB	85	21	4.96	1.30	7.77
siteC	86	21	5.08	1.98	10.20
siteD	86	21	5.68	2.40	12.64
siteE	86	21	4.93	1.36	7.60
siteF	86	21	4.82	1.69	9.45
Overall			**4.97**	1.76	12.64

Appendix C Landmark Error Analysis (CNN, All Sites)

Table 10: Mean Euclidean error per landmark (Approach 4 — CNN), averaged across all six sites. Landmarks are sorted from lowest to highest error. The dashed line separates the recommended selection threshold (≤ 5 mm) from the excluded set (>5 mm).

Landmark	**Error (mm)**	**Landmark**	**Error (mm)**
AC	0.06	mid_prim_sup	4.24
PC	0.93	l_prim_ext	4.51
rostrum	1.55	r_corp	4.83
mid_lat	1.63	l_temp_pole	4.87
mid_basel	2.01	l_corp	5.14
RP	2.14	*(excluded, >5 mm)*	
genu	2.21	dens_axis	5.66
RP_front	2.38	r_lat_ext	5.85
lat_right	2.52	r_temp_pole	6.05
SMV	2.56	l_lat_ext	6.12
lat_left	2.58	l_occ_pole	6.21
rostrum_front	2.95	l_front_pole	6.50
CM	2.98	right_lat_inner_ear	6.63
l_inner_corpus	3.01	r_occ_pole	6.98
left_cereb	3.06	r_front_pole	7.64
l_caud_head	3.12	left_lat_inner_ear	7.71
r_inner_corpus	3.20	top_right	8.02
mid_sup	3.22	l_sup_ext	8.03
optic_chiasm	3.23	top_left	8.15
r_caud_head	3.30	r_sup_ext	8.22
lat_ven_left	3.39	LE	11.23
mid_prim_inf	3.42	RE	12.56
VN4	3.46		
m_ax_sup	3.66		
lat_ven_right	3.67		
callosum_right	3.72		
BPons	4.04		
m_ax_inf	4.12		
r_prim_ext	4.19		
callosum_left	4.20		

Appendix D CNN Architecture and Training Details

Table 11: 3D CNN encoder architecture (Approach 4).

Block	**In ch.**	**Out ch.**	**Spatial stride**	**Output size**
Input (ACPC T1)	1	—	—	$1 \times 96 \times 101 \times 101$
Conv block 1 (stride-2)	1	24	2	$24 \times 48 \times 50 \times 50$
Conv block 2 (stride-2)	24	48	2	$48 \times 24 \times 25 \times 25$
Conv block 3 (stride-2)	48	96	2	$96 \times 12 \times 12 \times 12$
Conv block 4 (stride-2)	96	192	2	$192 \times 6 \times 6 \times 6$
Conv block 5 (stride-2)	192	384	2	$384 \times 3 \times 3 \times 3$
Adaptive avg pool	384	—	—	$384 \times 1 \times 1 \times 1$
FC1 + ReLU	384	256	—	256
FC2 + Tanh	256	153	—	153

Each conv block = Conv3d(stride=2) → BN → ReLU → Conv3d(stride=1) → BN → ReLU. Output 153 values = 51 landmarks × 3 coordinates, normalised to $[-1, +1]$ by dividing by `COORD_SCALE`= 120 mm.

Table 12: Training hyperparameters.

Parameter	**Value**
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$)
Learning rate schedule	Cosine annealing, $\eta_{\max} = 10^{-3}$, $\eta_{\min} = 10^{-5}$
Epochs	100
Batch size	4
Loss function	Smooth L1 (Huber loss, $\delta = 1.0$)
Augmentation	Intensity jitter ±5%; left-right flip with $p = 0.5$
Training subjects	≈516 (80% of all 6 sites combined)
Coordinate scale	120 mm
GPU	NVIDIA RTX 4000 Ada (20 GB VRAM)
Training time	≈45 minutes